

PROBLEMS-Simple Harmonic Motion

An object oscillates according to the formula $x(t) = 0.12 \cos(12t)$ where all numbers are in proper MKS units.

- (a) What is the period of the motion?
- (b) What is the maximum velocity of the motion?
- (c) What is the velocity when $x=0.10$ meters?
- (d) If this is the motion of a 0.100 kg mass on a spring, what is the spring constant?

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Most people hang their plants from a string or a chain (and some of them talk to the plants). You decide instead to hang your plant from a spring ($k = 8 \frac{\text{N}}{\text{m}}$) so that it can have fun oscillating up and down. After you hang your plant, you start it bouncing with an amplitude of 12 cm. You see that it oscillates with a frequency of 2 Hz.

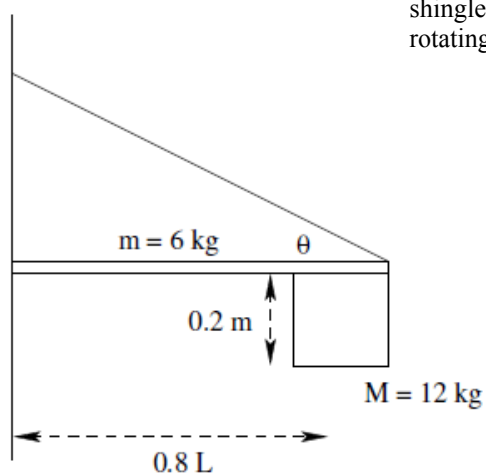
- (a) What is the period of your plant?
- (b) What is ω ?
- (c) What is the maximum velocity of the plant?
- (d) What is the maximum acceleration?
- (e) What is the velocity of the plant when its displacement from equilibrium is 4 cm?
- (f) What is the acceleration of the plant when its velocity is $\frac{1}{2}v_{max}$?
- (g) What amplitude of the oscillation would produce a maximum acceleration of $9.8 \frac{\text{m}}{\text{s}^2}$?
- (h) How far is the spring stretched when the plant is at equilibrium?

NOTE: When your plant's acceleration equals $9.8 \frac{\text{m}}{\text{s}^2}$ downward your plant feels weightless. However, you need not worry about vomit since the plant has no stomach.

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Now that you have your medical degree, you decide to “hang a shingle” in front of your home and begin your practice.

NOTE: Treat the shingle like a rod rotating about its end.



When a slight breeze blows, the shingle starts swinging with a small amplitude.

- (a) What is the period of the motion of the shingle?
- (b) If the maximum angle of the shingle is $\theta_{max} = 0.01 \text{ rad}$, what is its maximum angular acceleration, α ?
- (c) What is the angular velocity, ω_v , when the angle of the shingle is $\theta = 0.005 \text{ rad}$?